

ActInf Livestream #027.0 ~ "Active Inference: Applicability to Different Typ

Livestream | 1:28:03

three two one

hello and welcome to active inference

lab live stream number 27.0

it's august 23 2021 and we're going to

be discussing the paper active inference

applicability to different types of

social organization explained through

reference to industrial engineering and

quality management

so welcome to the active inference lab

we are a participatory online lab that

is communicating learning and practicing

applied active inference you can find us

at the links here on this slide

this is a recorded and archived live

stream so please provide us with

feedback so that we can improve on our

work

all backgrounds and perspectives are

welcome and we'll be following video

etiquette for livestreams

at the short link you can find the

roadmap of the upcoming discussions that

we're going to be having and these are

going to be group discussions on august

24th tomorrow and then august 31st for

number 27.1 and 2

and hopefully the author will be joining

us for both discussions and we have some

openings in the coming weeks if you want

to

join for a session or if you want to

recommend somebody who should join

today in active stream 27.0 we're going

to be learning and discussing about this

paper with the aforementioned long title
by stephen fox and the link is here
and just like a lot of
um the other zero videos all of them
this video is just an introduction for
some of the ideas it's not a review or a
final word it's kind of always this
assemblage of
all of the participants here just
modifying the slides and going down a
few avenues that we thought were
interesting and
setting up some questions to discuss in
the dot one and dot two
so we'll go over the aims and the claims
of the paper and the abstract and
roadmap and then talk about keywords
then go through the figures and tables
so if you want to participate in these
group discussions tomorrow or
the following week then just let us know
or save and submit your questions
and get in touch with us if you want to
participate
so we can just start with
hello and then it will be cool to hear
from both of you what
drew you to the paper or what made you
interested in talking about it
i'm daniel and i'm in california
so whoever wants to go first
i'm blue and i'm a independent research
consultant in new mexico and um i liked
this paper it was a really like
refreshing kind of interesting um and
very readable perspective compared to a
lot of the other papers and um it made
some links with

industrial engineering which i thought
was
were interesting and um kind of like a a
new take on active inference and fep
dean i'm i'm up here in calgary and
what i found interesting about the paper
is just looking at it through the lens
of
engineering's uh
engineering's
priors i guess in terms of
what they would hold up as active
inference just because of what they do
every single day
um and again
that kind of juxtaposed with the social
piece was really interesting to me
because
not to over generalize but when you're
focusing on how to break something down
you don't tend to tend to look at the
holes so i thought that was really
interesting uh to try to pull those two
things together and make them coherent
i agree there's that discussion of the
interfaces between the engineered system
and
the actual way it's used in the niche
which i think we're going to return to
when we're talking about like a
bookshelf
so the paper is
the title the link are shown here
and uh
i guess uh maybe
do either if you want to just read the
first uh two aims and claims
sure

so the principal contribution of this paper is to relate active inference theory to everyday practice in social organization

active inference is a cult

corollary of the free energy principle

which formalizes cognition in the

autopolitic organization of living

systems that resist the second law of

thermodynamics by occupying a limit

limited repertoire of states thus

persisting as bounded self-organizing

systems over time

okay and then i'll read the

last two then anyone can give a thought

it is explained throughout the paper how

industrial engineering and quality

management better enable social

organization to occupy a limited number

of states in order to persist as a

bounded self-organizing systems over

time rather than dissipating into their

surrounding environment

although previous papers by others have

encompassed active inference up to the

levels of sociocultural cognition this

is the first paper to address active

inference in everyday organizational

practice including industrial

engineering and quality management

so the paper is setting up to be just

like one more plank towards applying

active inference bridging some of the

practices that are already in use in

the fields that we're going to be

exploring like the quality management

and industrial engineering and then sort

of the adjacent possible for what kinds

of systems could exist in those domains

so

sometimes we have a more mathematical

paper or more

philosophical paper this one's going to

use a lot more of the jargon

and like dean said the priors of

engineering so that's why it's kind of

cool to hear about it

daniel can i add something to that yep

yeah i just think it's interesting that

i think we

to use a bridging metaphor is seems just

to be a really easy thing to do

and then i think what what the paper

in terms of the aims and the claims does

is tries to point out where that

that connection might exist

and we can always argue whether or not

it exists in all

situations at all times but i i do think

that there are some times again when

we can hold a static

connection between

an idea of one scale and an idea at

another scale and so

not all scales

allow for that or have those affordances

but sometimes they do so i mean that

part of it

maybe what we need to do is separate the

times when

the metaphor works and when it doesn't

agree being utilitarian implies some

sort of utility metric

having a preference for reward or for

preferred states implies those being

defined maybe and also i think this will

come up

in this paper

because it's all about which are the
useful metrics to check up on useful
policies to pursue

okay

so

does anyone want to read abstract one

i'll read it

active inference is a physics of life
process theory of perception action and
learning that's applicable to natural
and artificial agents in the paper
active inference theories related to
different types of practice

in social organization here the term
social organization

is used to clarify that this paper does
not encompass organization in biological
systems

rather the paper addresses active
inference and social organization that
utilizes industrial engineering
quality management and artificial
intelligence alongside human
intelligence

social organization referred to in this
paper

can be in private companies public
institutions other for-profit or
not-for-profit organizations and any
combination of them the relevance of
active inference theory is explained in
terms of variation free energy

prediction errors

generative models

and markup blankets

pretty interesting i think usually when

we talk about these multi-scale
perspectives on institutions cities
societies
it's always like
the entire edifice all of the niche and
everything that's moving in it is part
of this one biological organism
and
or at least biological co-regulating
system and here it's very explicit that
it's trimming the biological component
from the engineered component it's kind
of like the car and then the passenger
so then instead of just saying well the
car and the passengers are part of the
same extended cognitive niche because
that's one way to look at that situation
like it's a extended affordance that the
driver has
another way to think of it is by making
a very clean partitioning
recognizing that there's still this
immersion function like cars don't drive
themselves but you can still parse out
the individual who's using the car from
the design process of making the car
and one one thing that's added to that
if you haven't read
lucy and i'll always kill her last name
lucy suckman s-u-c-h-m-a
she's got a book out many years ago that
i think has still got some
incredible shelf life it's called um
it's reconfiguring the human machine
interface
and if you haven't read it it's a
fantastic book in terms of
looking at how

her case study is how people walk up to a photocopying machine for the first time where the engineers are trying to make it intuitive and how the human interface with that isn't always necessarily something that you can engineer for so something to consider if you're looking at this this if i could just give another thought i think that um the uncoupling of social organization from biological organization is kind of a really like unique and fresh perspective compared to some of the other papers that we've looked at i mean especially considering that like i mean it is biological organization because it's happening in in biological creatures but it's it's an interesting like partition that the author creates to um make it known that it's not talking about hierarchical organization in structures or societies yep so we don't need to worry about what's inside of the tissues and go into the biological organization okay i'll read the second abstract and then anyone can give a thought active inference theory is most relevant to the social organization of work that is highly repetitive by contrast there are more challenges

involved in applying active inference
theory for social organization of less
repetitive endeavors such as
one-of-a-kind projects
these challenges need to be addressed in
order for active inference to provide a
unifying framework for different types
of social organization employing human
and artificial intelligence

so i have a thought on that
it might be somewhat controversial but
um

you know

so

i think that um the highly repetitive so
i mean i think about active inference in
terms of exploration and exploitation
and i think like it really depends on on
where your generative model rests right
so like is the generative model like the
model of the organization or is it the
model of the individual because i i
think that um doing a highly repetitive
project that reminds me of exploitation
where like a one-off project can be like
exploration like so so you can still
like update your model depending on on
exploring some new avenue by this little
side project but like you come back to
updating your model the model of your
business yourself the way that you work
your organization

with knowledge about about how to do
processes in a one-off kind of process

i agree

especially because we talk about it a
lot in the context of generating
strategic novelties perhaps

and those are seemingly less repetitive

um but

i think there's gonna be um some further clarification on that

and also this points to this uh yeah

exciting areas like one shot learning or

like true structural novelty and active

inference models that's something that

we've talked about in other areas like

how do you know when to be tweaking the

knobs of the parameters as you currently

have them set up versus expand a hidden

state into two types of hidden state or

something like that

again thinking about what the system is

realistically or from this

instrumentalist way okay

the roadmap

is written out here

and it's a

we're gonna go through it in rough order

it introduces in each of these sections

one of the kind of key ideas or

principles the variational free energy

prediction errors generative models

markov blankets

and survival

and then in section seven active

inference is proposed as a unifying

framework so this is kind of one of

those classic paper layouts with some

disciplinary sections one through n

disciplinary sections

and then

uh the next section is active inference

as a unifying framework and then the

next section is like challenges or

current research directions and then a

concluding section maybe with a more um
call to action or more philosophical
uh liftoff
there were a bunch of keywords and i
think we're going to get to all of them
um so two columns of keywords
uh let me shrink the video
two columns secured okay
so
first
active inference
uh it's in the title of the paper and
here's how it's written in the paper who
wants to read the large text here
i got it
uh active inference is the physics of
life process theory of perception action
and learning
active inference generates predictions
active inference predictions are based
on knowledge learned from past
situations and perceptions of present
situations active inference minimizes
errors between actions that are planned
through the inference of predictions and
what happens when the planned actions
are taken
so this is the first lines of the paper
right in the introduction that's how it
begins with no
active inference is the first word so
that's sort of the opening shot and
it's just a cool
framing
to bring it out as a physics of life
process theory
yet this isn't about biological
organization so it's almost like

recognizing the origins
and some of the motivations for the kind
of modeling that active inference
provides
yet making another later decision to
partition the system of interest a
certain way so cool framing and then the
uh next
lines so kind of shrinking that
definition of active inference
um
this is because prediction errors
cause unwanted surprise
surprises which individually or
accumulatively can threaten survival by
going beyond the limited number of
states in which survival is possible by
contrast minimizing prediction errors
facilitates survival through least
action
so this is also linking up several of
those key terms that are going to be
section headers later but it's just this
idea that um
predictions about the external world
especially if they incorporate
preferences that are optimistic
help reduce the complexity of deciding
which actions to take and then the way
that that gets operationalized is with
the process theory of prediction error
minimization which is
a real-time estimate of how good the
generative model is performing
and then just
add anything or
say more here but
it's just an interesting

topic how it's described as the
principle of survival through least
action
so
what is this principle of least action
and
how do we reconcile
active biological agents
as being in one way participating under
this survival through least action even
if it's just a misnomer what does that
really mean
and then how does survival through least
action
contrast or overlap with
more familiar phrasing probably survival
of the fittest like
is the survival of the fittest the
survival of the least
active or how should we read this
relationship between
uh maybe darwinian conceptions of
fitness and which individuals survive
versus this physics based
framing of survival through at least
action under the
principles of least action
stationary action principle that's
something daniel yeah
yeah so i think it's interesting that
you
depending on how you define each of
these terms
that you can you can sort of
transfer or oscillate across frame what
a frame is which to me is
identifying a focal point maybe a margin
or a limit and what if it is what goes

inside the other like
glove and hand or hand and glove and i
think sometimes we talk about
a niche fit
but that's that's different than a what
the frame of the niche is
so the fact that we're we're sort of
ping-ponging back and forth
between the two i just wonder whether or
not
we're looking at the same thing or if we
realize we're looking at two things at
once
nice
continuing on with the introduction
it's written
active inference is a corollary of the
free energy principle fep which
formalizes cognition of the autopoietic
organization of living systems
within fep active systems must occupy a
limited repertoire of states this
requires minimizing the long-term
average of surprise associated with
sensory exchanges with the world
minimizing surprise enables them to
resist a natural tendency toward
disorder
so
this just kind of relates to some
questions like
how do framings for this sort of active
intelligent behavior
what other framings exist
and how do we think about
broad concepts like reward preference
and curiosity and surprise in active
inference

um and then the last paragraph then any thoughts or not surprise rests on predictions about sensations which depend on an internal generative model of the world

in particular although surprise cannot be measured directly a free energy bound on surprise can be suggesting that agents minimize free energy by changing their predictions about what sensory inputs will come from actions or by changing the predicted sensory inputs through changing action

this just reminds me of the kind of fundamental uh

pairing in active inference where the ways that you can go down that free energy gradient are either by updating parameters of your internal model learning development or by updating action states

which is through action because you can't control external states directly and you can't control the incoming sensory data directly

then all you can do is control those two states that are under your control the internal states and action states

all right another term was the markov blanket

so anyone want to just like

give what what's what's the markov uh 101 or what's

what's the way in which mark-up blankets came into play in this paper or in another context

so

um marco blankets are essentially the

the boundary or or how how a system
creates a boundary and how you can
define essentially like a
non-equilibrium steady state system with
a martal blanket um i thought it was an
interesting

um play in this paper and i think maybe
we'll unpack it here in later on in the
in the intro video

we just in number 26 talked a bunch
about markov blankets so suffice to say
that there's um

it's still open exactly which cases
which sense of the markov blanket is
meant on this continuum from analytical
and mathematical with markov og blanket
the bayesian statistical generalization
by pearl and others and then this
partitioning into incoming sensory and
outgoing active states of the blanket by
fristen and others and then how that
exactly relates to all these other
topics that we're
discussing

okay i'll play the video at um 46
seconds blue what was exciting to you
about this

uh so this is the video um from the
tesla ai day that just happened last
week and this is um

it's a humanoid um
robot that uh tesla is going to attempt
to design

um and so i just pulled this definition
of artificial intelligence from um
like google and it said the theory and
development of computer systems able to
perform tasks that normally require

human intelligence such as visual perception speech recognition decision making and translation between languages and and so spoiler alert uh this is actually not uh not a robot this is actually like a person um but but he made it you know look like a robot that was getting up and then like bam i'm gonna dance but i think like that kind of movement and perception about about space that kind of spatial awareness is something that's still very much a challenge for um ai so so that's why i threw this video in there agreed will be interesting to see how they do processing and what kinds of algorithms and hardware and software end up being used for real-time robotics systems any comments on artificial intelligence or or anything dean again i don't spend a whole lot of time on this because to me artificial intelligence is always a it's always a post-hoc domain even this example um can we create a copy of this so i mean it's already been done by something and now we're trying to re-engineer it or re-develop it as opposed to what active inference is which is the inclusion of ad-hoc the what if so i mean if they can get artificial intelligence to a place that it's it's including what if in that in the truest sense of all possibilities then

i'll probably

be quite excited or scared i'm not sure

which

nice

so

some other keywords were social and

organization

and one of the

goal statements of the paper was

this paper is intended to contribute to

bridging the gap between theoretical

papers in active inference theory which

have been reported to be too opaque to

be understood widely

and potential application opportunities

in the multi-intelligence social

organization

that being social organization that

employs human intelligence and different

types of artificial intelligence

so

and whoever put this image what did it

mean or represent in the context of the

paper

so i i just looked so i didn't

i don't know that it's necessarily

really in the context of the paper but i

i started to look into like how are

societies organized right and so this is

like i just was doing some googling and

i thought that this was an interesting

um paper and the title and the author

there in the small print if you're able

to read them but

this uh is like a um a state

organization right and and it's you know

describes the fascist media model model

as focusing on community and culture

and it describes the communist uh media model as focusing on state bureaucracy and the liberal media model as focusing on market exchange which i just i just thought was interesting and just kind of an interesting way to um to

i don't know reflect on societal organization and what what is the central theme there

and it's uh kind of in line with that partitioning of the kind of the car from the user because not included here are like the human users this is kind of everything but the eyes on the tv this is just partitioning the the structure of the organization as an engineered object

so that's a cool um similarity

yeah

so we just had a guest stream and and pranav and raphael

even they were debating

can we can we say that something that's an emergent property in an individual which is then something that we can test as an emerging property within a dyad can also become an emerging property within these high variability states and even they couldn't agree

that there's there's some sort of a parallel

i think we know that on these high variability states like a social organization

afterwards again we can

we can take an account of what it became
but we it's really hard to know
where it's going even if you break it
down into a
communist or or a fascist
direction it's telling us like you say
what what direction the high down the
highway the car is going but that but
beyond that it's still a lot of guessing
yep
so i guess that's where industrial
engineering comes into play for that
designing for that car on the highway
industrial engineering involves applying
methods such as task analysis and job
design
in order to predict results evaluate
results and improve results from
processes during their development
the application of industrial
engineering has been progressing around
the world since the 1900s
so
wasn't an area i was too familiar with i
saw a lot of mention of the 99 out of
100 and the three standard deviations
and then halt the process
so that kind of just was an interesting
uh thing to hear about
any thoughts or we can continue on
okay as i guess a related field of
industrial engineering is quality
management
and so this is a type of system
that involves i guess hardware and
software and behavioral protocols
quality management systems qms involve
documenting processes which have been

developed through industrial engineering
as process specifications work
procedures etc
monitoring processes for conformance to
specifications etc and using
observations of non-conformances to
inform the further development of
processes both industrial engineering
and qms are focused on the continuous
improvement of processes the application
of industrial engineering has been
progressing around the world since the
1900s in quality management since the
1950s

any thoughts on that

just having been a business owner it's

uh you know it's it's a process

yep definitely makes me think about uh

precision of measurement changing and
then

what can be measured and what can be
measured gets entrapped and then what
can be regulated becomes gamed and all
these sort of complex ways that
assessment of performance

fits one into local versus

global optima

you can also see how instructionalism
got systematized

yep if there's a way to do it and this
is the way that's approved

um even if the later becomes like a you
know an improvement in the protocol or
some sort of uh change in the context
maybe it can't be utilized because it's
not part of the approved version
and so

there's no simple answer there because

you can't just say well okay accept all changes and then if you just say okay we'll have a small bar for making edits it's like there's no perfect trade-off point with the false positives and the false negatives especially for situations that truly are ambiguous which is why we fall back to these approaches that put us at the kind of functional inflection point of the pragmatic and epistemic outcomes which is like active inference rather than just pursuing reward over short term or over some other time frame but common to all these different frameworks for control theory is basically statistical process control so maybe whoever made this slide was it yeah blue go for it so statistical process control is a collection of tools and techniques used to measure and analyze process data in order to characterize the behavior of our processes and achieve process control and so in the paper i don't know how much we'll get into it here but the paper really describes statistical process control and gives a figure similar to this so you have normal process variation which they described as six sigma uh deviation from whatever is optimal so there's the optimal which is like the green line and then three sigma

to the up and three sigmas to the down
as to like an over or under estimate or
of your you know target
and then anytime that you have a process
that deviates from the six sigma range
you um have this special cause variation
and and so like in like
the business

schema that i'm familiar with at least
which is minimal but some um you know
you have you do this root cause analysis
like why is there this special cause
variation like what is what happened in
that um in that one circumstance we have
to go and analyze that instance to find
out why it was outside of the sigma
deviation because it's just like that's
an unacceptable data point at that point
nice good explanation

how about thermodynamic entropy
uh also um

this was also me but um it's a quantity
representing the unavailability of a
system's thermal energy for conversion
into mechanical work often interpreted
as the degree of disorder or randomness
in the system

and so you know there's the law of
thermodynamics that says that entropy in
the universe is always increasing so
um becoming going from
more ordered to less ordered or so from
highly ordered to more disordered which
this kind of shows in this example uh
things spreading apart and like it's
like you would if you left town for two
years and you came back and found that
your house was a mess that wouldn't

really be surprising like even if you
left it clean even if you know i mean
just things deteriorate and fall apart
and dust goes in and whatever rodents
creep around and stuff like that
and it's that apparent
and empirical local organization of
matter like into bones and flesh and
life and whatnot
and also the always and only
co-occurrence of that kind of local
organization of energy with cybernetic
or anticipatory behavior
that's what motivates a kind of
framework that is going to on one hand
work for particles but also work for
active particles and all of the
continuum between so that's why there's
all these different cognitive structures
as well as physical structures that are
being linked together through basically
physics
and by hooking to free energy principle
and variational methods and all these
other areas of math and physics then it
puts active inference just like one step
closer towards embodiment on some of
those different
pathways
okay
information theoretic entropy
uh so this is the the shannon entropy as
it's often referred to because it was i
think first described by claudes shannon
um but i always put uh maxwell's demon
here when i'm thinking about relating
thermodynamic energy and informational
entropy and i think was it alex kiefer

wrote a paper that really
we did in one of the live streams that
really kind of outlined how these are
connected with the helmholtz
distribution and so if you really want
to dig deeper into these relationships
um that's a good way to do it but i
always think about you know so maxwell's
demon is is the idea that you know if
you have molecules like shown on the
left here in you know this box and you
have two sides of the box and there's a
partition
in the center and like the you know
they're cold and hot molecules
so if you knew which molecule was hot
and which molecule was cold so the demon
is sitting there who can see is the
molecule hot or cold and only lets the
hot ones go through to the right and
only lets the cold ones go through to
the left and every time he opens the
partition
effectively that is is work
and so just kind of conceptually um
that like shows that that the
relationship between information and
work the more information you have
the less work you have to do maybe so
that's kind of the relationship that i
think about in my mind
and then there's a formal definition of
shannon entropy um down below but it's
usually
given in informational context as a
measure of surprise as we've talked
about on the live stream many times and
they relate information as negative

entropy so it's the the reverse
surprises the reverse of uh surprise or
sorry information is the reverse of
surprise
cool thanks for the explanation dean any
thoughts on that
yeah one of the things i i mean i love
this thanks blue because thank you for
that explanation it's really it's really
good so one of the people i'm reading
right now is a
guy by the name of dan siegel and his
book about the mind and what he talks
about is
um when people are at dis ease so not in
an easy state but when they're when
they're struggling with something he
talks about the fact that there's one of
two ends of a continuum be it blue and
red
in this example one one end is chaos and
the other is
rigidity
and so from an information theoretic
entropy state are we stuck are we bored
are we feeling trapped or are we being
overwhelmed or underwhelmed
meaning we're we're
we're not in that in between state and
so i think that
i think there's a lot there's a lot that
can be done if we get a basic understand
understanding of maxwell's demon without
sort of getting
scared off by it
cool it's almost like um yeah what would
these particles and their temperature
represent if they were i don't know

ideas or something more related to
inference rather than just particles
because we're always looking at that
intersection between the physical
systems and the informational systems
all right

variational free energy

so this keyword was defined in the paper
as the difference between predicted
sensory inputs and actual sensory inputs
when monitoring processes

um

vfe increases as the difference
increases between what is predicted to
happen if actions are taken and what
does not happen what does happen when those
actions are taken the more prediction
errors there are the more thermodynamic
energy will be consumed amidst the
thermodynamic entropy of remedial
actions corrective actions restorative
actions and firefighting this will leave
less thermodynamic energy available for
doing productive work comprising
productive actions

thus if the variational free energy is
high then thermodynamic entropy eg
energy not available for doing
productive work will be high and the
thermodynamic free energy
eg energy available for doing productive
work will be low

hence variational free energy can have
an inverse relationship

to thermodynamic free energy

this is because the bigger a prediction
error which can be considered in terms
of bigger vfe the smaller can be the

thermodynamic free energy available to
do productive work so this is a pretty
interesting
passage did either of you have a thought
on it
um
it's exploring this thermo thermo info
nexus in a little bit of a different way
and
on one hand it's kind of confusing that
the variational free energy
is going to be the opposite of the
thermodynamic free energy but it's
almost like
what is happening here and um it just
to see that they have a negative
relationship it's like if the engine
were perfectly anticipatory and super
efficient it would be carrying out a ton
of work if it's moving in a super
spurious heat producing way
then the product a lot of variational
free energy a lot of surprising
observations
relative to functional uh operation of
the motor then your capacity to perform
work
to capture those very noisy vibrations
is low and so a lot of the
underlying
uh potential becomes dissipated rather
than used to perform kind of useful
inference useful action so not sure if
that's a correct
mapping but i think it'll be cool to
talk to the author because this is a
very fascinating idea about how
different

formalizations of free energy um
actually relate to modeling processes
yeah blue
so it's like i i really thought this was
kind of what made me think of maxwell's
demon right so
it was interesting because you know the
variational free energy when we talk
about minimizing free energy we always
think about like as minimizing surprise
right so if we if we minimize surprise
then then we don't have to do all this
thermodynamic energy like the
firefighting that they talked about and
i think that the paper when we get into
the five wise and that figure i think
we'll elaborate a little bit more on
that but um

the
it's just interesting because we don't
have to output as much work and that's
kind of what made me think about about
maxwell's demon and kind of why i snuck
that in there um but yeah you're right i
don't know like you know i think that
the relationship between
um

info dynamics and thermodynamics is
still very much being defined there's a
lot of very recent research um that's
looking closely at how those two uh
relate to one another

nice dean

yeah i just

i immediately

said to myself

i really want my autonomic nervous
system to work and i don't want to have

to think about it because
a lot of the mechanical things
are dependent on that so
i'm not
i'm not sort of pulling apart what the
claim is here
but i think
if you read that you're kind of saying
okay
um
[Music]
a lot of the surprise you're not even
aware of because you're on an autonomic
nervous system is functioning correctly
great
point there it's almost like a surprise
variational surprise does not equate to
attention or even especially the
conscious awareness of
attention in a human
psychological situation
so it's totally possible that like the
difference in pressure sensing depending
on your body's posture that's
completely normalized out that's
recalibrated for the expectations of
body position
without coming to our attention unless
we bring it to our attention so that and
many other similar examples
so dean i love how you're like not able
to to stay outside of the biological
partition
right
well i i just i'm not again i'm not
i want to be able to create with this
stuff
i mentioned that and so

to me

to willfully ignore that is hard for me

to do i'm not saying it's not possible

i'm not saying it's it's the right thing

to do here i'm just struggling to let go

of it i want to let go of it but i'm

having a hard time with it well and i

think for all of us you know the urge to

apply

to every system is very strong

right here's table one so here's the

mapping

of a bunch of different sets of terms

from active inference constructs

and then thinking about how they're

applied in industrial engineering and

quality management practice so

variational free energy which we just

we're talking about

is the difference between the predicted

sensory inputs and actual sensory inputs

the variational free energy upper bound

as well as its lower bound is basically

the range in which that system is

working well it's like the loaves of

bread are within the calibration range

and we can just keep on pumping them out

precision waiting

has to do with which aspects of the

process are monitored with which what

level of precision

for example which width of a part has to

be machined in fractions of a millimeter

versus other tolerances

and then the expected free energy is the

expected energy

between the preferred sensory inputs

from process operation

and the actual sensory inputs
so the variational free energy is the
difference between the prediction and
the actual so you have some generative
model of your data center and then you
have the generative model of the
temperature through time
and then the comparison with the actual
but here it's about the preferred
versus the actual
so that's like we we we recognize that
it is warmer than we would like it to be
which is why we're conditioning our
policy in the long range towards getting
back to that zone but for now we need to
also have an accurate sensor measurement
so that's also a nice juxtaposition
that's not always finessed in other
papers
the next and all these keywords the
first set were about free energy and
precision and how these terms are
calculated relative to one another
the second set of terms in table one
is some of the
key terms of bayesian statistics
so just to read only the constructs um
and these are also really helpful
because
we're as a lab thinking about what are
the core terms of active inference what
are the definitions that are general
enough to be
really
educational but also not too limiting or
prioritize one sense over another for
example so the bayesian terms that would
be found in any other course or textbook

on bayesian statistics

prior expected posterior actual

posterior preferred posterior and

prediction error

so that kind of puts the predictive

processing as a

under the umbrella of bayesian

statistics

okay

the next set of terms in the table

was related to this dichotomy of two

different types of inference so

epistemic inference was defined as

updating the definitions of process

during process improvement based upon

analysis of non-conformance reports

whereas instrumental inference is

updating which actions will be included

in the process during process

improvement

so this is sort of like epistemic

knowledge based this is about updating

the knowledge

based upon especially the times where

things don't go right

and then this is like

policy inference action planning as

inference because it's like

instrumentally is actually like about

what the instrument is going to do

so

and then the last

okay any thoughts on these terms

okay then the next set of terms in table

one

was

mark hall blanket

internal state and external state

so definitely check out number 14

number 26 a few of the other live

streams to learn more about the markup

blanket and this is the figure from

uh decosta at all 2021.

the blanket

is here the boundary state between the

internal state

of the social organization

so those are the ones

and the and the external state of the

environment when operating processes are

defined in the quality

management system

then what are those internal states

those are the social or organizational

processes developed during industrial

engineering for minimum risk in the

process environment

and then the external state is the

environment

so

that's kind of mapping the markov

blanket kind of the car

the interfaces and outside the car or

maybe that's not the perfect mapping but

i think that's kind of what is being

suggested

and then the last

few terms of table one

are four of just the general constructs

general principles that are also section

headers and important topics risk

ambiguity principle of least action and

principle of survival

dean

yeah so i think all of this ties in

to a

a way of looking at the world through
through rows and columns which is fine i
mean it's it's a good it's a good place
to try to look at things and analyze
them but
not everybody
not everybody creates off of that
because not everybody
is necessarily looking for a precision
forward view sometimes
you're you're experimenting and you're
making all kinds of mistakes and
learning from that as well and so you
again you just want to just want to see
this for what it is it's it's nothing
wrong with it but it's not
it's not the only way
that's all
blue anything
okay so then
there's the figures
and these will recall some of the key
terms we talked about earlier
uh the text says as shown in figure one
spc charts these statistical process
control charts
have upper and lower control limits
and processes need to remain within
these limits processes that go outside
the limits do not meet specifications
that define intended states so left side
good top of the drake meme right side
bad bottom of the drake meme not working
okay
figure two is this five whys so it's
written out here more textually if you
wanna pause it and look at it and here
it's more graphical so do either if you

wanna
give a thought on the five wise or
okay well the scenario that's laid out
is uh
kind of working backwards
from why there was a not reliable
organization
and the five wise it's actually not
related to aristotle's four y's but that
would have been a nice opportunity maybe
there's a four by five matrix but these
five y's are referring to
um needing to go back kind of five
layers
or like five
derivatives up the chain
asking
what led to a high reliability
organization being unreliable
so the general case of this sort of
cascading failure
in terms of variational free energy so
the general phrasing is that there's
high
uncertainty high entropy in work
instructions so the ikea instructions
are not accessible enough to many people
or the radio is really blurry so people
can't hear what letters are being said
and then that makes the variational free
energy high because of prediction errors
which then leads to unproductive actions
so then you're you're scanning on the
radio because you can't quite lock on
and then that's consuming energy
and then that ends up reducing your
battery so that you can't do what you
actually need to do and then the case in

this truck driver incident
is that the truck driver is getting
ambiguous
logistical data so this itself is a
policy um communication from like you
know some cloud mapping service or the
company that's contracting the trucker
and then this leads to
the driver having increased spatial
uncertainty getting lost
which leads them to drive around
unproductively so they don't have enough
fuel to efficiently complete the
deliveries that are important
and so here's the graphical layout of
the five wise
showing with that example
how on the bottom is the truck drivers
getting the bad
info
and then that surfaces
after sort of several repercussions in
terms of the uh
the um structure of multi-level decision
making
and active inference
how we can
use this framework which maybe is used
in industry
and map it onto the states of decision
making from active inference perspective
but notice that there's no reward here
it's not well drivers
minimize their reward about this or you
know risk on this or maximize their
reward on this so it just shows how this
sort of um framework
has a different framing than people

might expect if the only machine
learning model they were aware of was
machine learning and
specifically reward reinforcement
learning dean
yeah and it isn't just industrial you
can
you can be like me and
jump in the car with two of my friends
who are both
emergency doctors and you're heading off
to the ski hill and at some point the
conversation comes around to there are
no accidents in the world
and of course being emergency doctors
they see the the
outcomes of
of
events
happening to people and so their
perspective is there's always a way of
being able to go back and see
all the
parts of the decision branch and how it
arrived in front of them and them having
to sort of put the pieces back together
again but
again i don't
i don't know that necessarily
that that means that randomness doesn't
come into play and there isn't a
certain part of this where the variation
simply can't be
um
[Music]
calculated in but
i still go skiing with them i don't i
don't

dislike them or defriend them so anyway

yeah

um

it just reminds me of like

when i dropped a hard drive then i

thought well i guess this is the 100

hard drives that fails every year

yes but it was also the one time that i

dropped it so there's always this sort

of like n equals one way to uncouple

uh cause and effect or potentially uh

actionable consequence to the not

reliability of your organization but

then also there's this designing for the

statistics when they decide to make a

recall or not based upon you know the

relative amount of failure

versus

the relative amount of you know legal uh

changes needed something like that so

that's where you talked earlier about

including co-occurrence instead of just

only giving time over to the sequential

aspect of this

yep

so figure 3

frames the quality management systems

the qms the quality manual as a

generative model comprising beliefs

about processes

so it's interesting because it could be

seen as just a cookbook recipe book

instruction book

but it's being

at least deployed in a generative

capacity

and instead of just like saying you know

whip the eggs for five minutes and then

um until they're

frothy or you know until they're this

make peaks it's like we expect it to

make peaks

whip it until your expectation is

achieved

it's a different way to get to that same

outcome now which one is going to be

more resilient if it says five minutes

and there's no peaks then what do you do

do you add something do you keep on

going whereas if you have a precision uh

seeking

process maybe you can be resilient to it

if it only takes two minutes and you

don't overdo it

uh and the the example of something

being correct like 99 out of 100 times

is just used i think just repeatedly in

the paper

just

it could be calibrated differently and

i'm sure it's situational it's just sort

of like shorthand 99 out of 100.

um but yeah

it is defining these qms

in terms of uh the actual active states

so what should be done you know pipette

50 microliters of fluid into the little

tube and then you monitor something

about the process

and then that

leads to this feedback cycle between the

uh the quality management review

and then that

being

in kind of co-evolution

with the industrial engineering itself

and then that leads to the update of the process in the quality manual in step four

so that then there's kind of like okay if it's too low after your first pipette then do a little bit more pipetting and then they keep on protecting that 99 um

functional outcome at each of those nodes that are bifurcating out okay any thoughts

uh not gonna

read this whole thing but

the top part talks about this design for assembly and one of the principles is to make parts either symmetrical or clearly asymmetrical so nothing in the uncanny valley this is done so that it is immediately and unequivocally apparent how a part is to be put into subassembly and the this is the famous uh ikea catalog aesthetic

and how that made uh furniture building like fun and accessible for some people i guess

and then in this

second quote

it is kind of thinking about that ikea paradigm although furniture companies may not be considering their industrial engineering in terms of minimizing variational free energy it is apparent that its aim is to minimize errors between what it intends to people to see when looking at its furniture kits and what people do see when looking at its furniture kits

so it's sort of like it's

the goal should be to make it look like
it looks on the sale room floor or like
it looks like in your imagination after
you see the advertising
and then if you can see those flat
pieces of wood and you can see that
instruction booklet and you can see the
policy

then you're going to be not surprised
you're going to be satisfied because
your preference was to have like the
couch there

so just kind of a cool example that
highlights how

our preferences aren't just over
temperature

we often talk about those kinds of
examples but we can also have
preferences for

having affordances or niche

modifications like we can have a
preference to have a handle on something

and then that kind of relates us to this
like behavioral economics under active
inference like what makes somebody
expect there to be a new couch in their
living room and then what makes them
drive out and pick it up on the weekend
and everything

and then

um

it's this kind of idea that the supply
chain is

this extended uncertainty reducer

and it's like bringing stuff to you

and then there's this notion in this
computational industrial case and in the
at home case

where

the assembly information that's provided is a policy decision from the people who are making the product and then that really relates to like participation in making that product if it's all pre-compiled then there's no affordance for participation versus having like a culture of people working through it from many different ways so and then this was just the funny example of the interface

which said uh

although there may be little risk in furniture assembly this can be followed by much ambiguity and furniture installation so in other words it's not just about what's on this brochure because pre-determining all the possible prediction errors in fixing the assembled furniture to walls in millions of different buildings around the world is very challenging and so there's this difference between the assembly information like how do you just assemble the system of interest uh which is just the internal and blanket states of the new uh shelf what's the difference between that information how to build the bookshelf and then the operations like oh well you didn't tell me that i couldn't kick it from the side it's like you just told me how to build it you didn't say that i couldn't use it this way and then so how do you design for internal coherence versus designing for interoperability

with sort of unknown unknowns and all
these different types of entities that
the system's going to interact with
on the blanket on the interface
so it's not just the ikea
booklet but i thought that was kind of
an interesting notion that even just
building it wasn't really the end of the
story because then people take their
perfectly built bookshelf and then just
like staple it to the wall the wrong way
like ruins both um so dean
yeah and then
and i don't know if you can see this but
then how do you account for this how do
you accomplish somebody who
took a piece of olive wood
any random piece of olive wood and
turned it into something
like this right so it's not there's no
there's no billy there's no way of
deconstructing this you just have to
pick up
you have to pick up something that is a
mark off blanket it's it's it has it's
pre-carved
and then you have to be able to pull out
what you think is in that and how do you
how do you engineer that
well it's it's a completely different
process so
i think i think we're i think as people
were capable of doing both
and i and i just i'd love to talk to the
author
about how how you
how you
do this kind of work when

there's certain results that are
are available and are part of the active
inference model so
it'll be a nice contrast tomorrow yeah
yep nice good point like where's artisan
ship and uh craft in this phrasing
so uh here's an example using that uh
case of the like bookshelf
or the cupboard being bolted to a wall
and this also gets at that partition
between the part of the system that you
can design
and then the part of the system that is
not being considered
so
here the internal states and this is
kind of interesting
it'll be cool to see who is there live
to talk about this sort of embodiment
aspect
the internal state is the person's
embodied prior knowledge of fixing
assembled cupboards to walls and
depending on what they know about you
know carpentry if they're just totally
watching the youtube video and then uh
letting it rip versus if they're
experienced
they're gonna with different materials
they're gonna take actions
and that's going to modify the niche
which is the external state the room to
which the cupboard will be fixed
and that's going to
be reflected in what the person
perceives
and then it's just cool like the
here

maybe there's multiple ways to interpret
the center

it's like there's the wall blocking like
external blocking state then there's
like the affordance block where the
tools are inadequate like you know you
need the hex wrench of size seven and
you only have the size four so you just
can't do it even if you know how to do
it and there's times where even if you
have the right tool it's like it's too
rusty to do it

so there's like it's just a cool
example

and
shows a little bit of that that
horseshoe theory again between
philosophy and engineering where they're
both radically focused on embodiment and
pragmatics

okay

then

the next section after four is seven
active inference as a unifying framework
and so basically they write
an important application of active
inference theory can be as a unifying
framework for a social organization
framework summarized in figure five
which we'll look at in a second
operated through quality management
systems qms via interoperability
protocols at an upper level of
multi-tier enterprise architecture so
before we look at figure five

it's just

like how does active inference serve as
a unifying framework in some of these

areas that we've been talking about for the last months like we've heard people use this integrating word to talk about active in action and perception engineering biology communication cyber physical systems remote teams etc so just it's being used a little bit in a different way here so let's see figure five and then you can give any thoughts on it here organizational active inference is again being mapped onto this uh sense internal action external loop this kind of circular loop the internal states are the organization's belief just like in figure four it was the person's embodied beliefs here it's the organization's embodied beliefs embodied in the qms which is again being framed as beliefs over uh processes and states that results in human and artificial intelligence employed to fulfill specifications procedures etc so these can be actions that are taken by apis or bots or by people this acts upon the external states which lead to perceptions of sensory states which could be the temperature in the room it could be the um the ph of the wine it could be the number of sales on the course

so then they do industrial engineering
on this loop as an organization
in relationship to integrating
perception and action through internal
state updating pretty much just like
we've been talking about for all these
other active inference systems any
thoughts

okay

table two

uh writes there are many there can be
many reasons why pragmatic subsumes
semantics and leads to prediction errors
for example inept priors and inept
preferred posteriors can come from
lock-ins path dependencies and success
traps which place emphasis on past
experiences over new information
individuals may seek to survive by
learning from filtered information that
supports the preconception of groups
which they believe can support their
individual survival rather than seek
information that supports the efficient
reduction of prediction errors in
external states

so it's almost like

you can't prefer what you prefer so much
otherwise you're gonna um i don't know
maybe follow that local band all the way
to a local end

instead of sometimes

coming out of that local

uh local local attractor

and uh any

want to go for this blue yeah yeah just
it seemed like just confirmation bias
right you seek to survive by learning

the the information that supports the
preconceptions of groups right so that's
like you know i believe this my my can
can my group believe this like i don't
know that's that yeah this is a cool
table it's like okay so in the
perception phase of active inference
you can have a top-down error based upon
inept priors
so you can believe the wrong thing about
what you should expect to see or
you can have the sensor be influenced
interesting about eg emotion but um
that's i guess being phrased as a
bottom-up way that
perception can be modified
and then in the intermediary
intermediate it's bounded rationality so
not that's it's kind of like a
compromise between what you think is
possible and
magical realism
then the action phase also can be uh run
um a ground by inept
uh
focusing on basically poor
outcomes like if you're
aiming for five degrees off of shooting
a free throw then you're gonna miss the
free throws
and then that also your attention
could be affected by poor signal to
noise ratio like if you couldn't see
where the hoop was
and just interesting how like it's being
really methodically broken down the
sources of where the error can come from
and then also

um in table three there's a nice distinction and it relates to that notion that active inference would be uh potentially very effective for repetitive processes but potentially not as effective for one-of-a-kind processes and so the repetitive processes are like mass production and then the one-of-a-kind processes are like the project production so right off the bat it's kind of like well how individual does a project need to be before it's not uh best modeled by active inference and then it just talks about how markup like it's generative models prediction error and variational free energy play out in both of these two areas so that would definitely be something for people to like read through the table and then mention it and give a specific example or question to the author because this is like super informative and a cool combinatoric way to look at how ideas are related so yeah dean and then blue so when i'm looking at this and i don't know if this is helpful or if this just makes it even more turbid but if on one hand of this is the ability to frame something so that we know what's in what we're looking at versus what we're not looking at and if the other hand is how we match essentially if you go back to

table two and this table three and and
the figure the matching is what's super
ordinate and what's subordinate what's
inside of what
those are the two hands that still
doesn't tell us what is going on in
between which i think is the fit piece
which from uh
from a thinking standpoint is am i
fitting now
hand in glove meaning
is is my is my glove size size large
or is it glove in hand does that does
the glove i need
serve the purpose because i'm skiing or
because i'm i'm
i'm foraging or i'm or i'm water skiing
right so
between
framing
and
and matching
is this fit question there all three of
them are different things the fit is in
between these other two things and even
if we
make a great table
and even if we make sure that we got the
right headings and right definitions
underneath it still doesn't talk about
is it
glove in hand or hand in glove and i
think for people who want to actually
use active inference
they want to be able to not get stuck
conflating
fit with frame
or

complaining fit with match they want to
put those two parameters the the frame
and the match on the outside and then
see what happens in the middle based on
what those parameters are i think that's
the struggle for most people
trying to gain entree into well
how does this active inference thing
actually work
it works different all the time that's
the point of the markov blanket it's the
potential
to be able to invert
at will that i think
laying it out this way helps it gets you
like 90 of the way there but it doesn't
really tell you what you could do with
it as opposed to
what just happened
very interesting dean
blue
so
like just kind of this piggy backs right
off of what what dean was saying like
the markov blanket and even here in the
first um
the line of this table
um
you know the markup blankets are hard
when they're shifting right and this is
something that we've talked about in
other um live streams in other contexts
like shifting overlapping my markup
blanket interacting with your marco
blanket i think that um you know the
underlying mathematics to support those
kinds of
um arguments are kind of missing still

or or not not missing but still being
worked out is probably more correct
um but but i do think that markov
blankets do change i mean even if i just
think about like my brain as you know
markov blanket like cells die cells are
born and i know that that's very
controversial in the neuroscience field
but but cells are actually born
so so i mean there's there's constantly
this this turnover of who belongs inside
the markov blanket and any kind of you
know organizational structure has that
shift it's not like the boundary of the
markov blanket it has to be has to
retain some kind of flexibility and and
so that um you know is is seen here in
this in this um
first line um and also who's generative
model is it
like so this is kind of another
something that that i was like
so the quality management solution or or
the qms right quality
i maybe got the abbreviation wrong but
um
that that is the gen that is the
generative model but like who does it
belong to is that does it belong to the
process does it belong to and this is
maybe my like a deficiency in
my understanding of
industrial engineering because i'm just
not familiar with the jargon but like is
it the process that gets the quality
management solution or is it the
organization that has the quality
management or is it like some kind of

sub unit is it a team so i'm just kind of unsure about they talk about this is the generative model but like whose model is it and like so depending on on how that's answered and we'll have the author on so we can ask them but depending on how that's answered you know the these one-of-a-kind processes as i said earlier can maybe you be used to update the generative model in a very um explorative kind of way

nice

so here's just for a few

uh implications and bigger questions

so here's just a quote markov blankets

are not features of real world systems

rather they are an intuitive post-hoc

ascriptions

rather they are intuitive post-hoc

scriptions made in order to model

real-world systems

hence any description of a markov

blanket is subjective and can be changed

over time as the modeler's focus changes

and or as real-world systems change in

the opinion of the modeler

thus as summarized by phrases such as

the map is not the territory markov

blankets

and then also it this is a funny line

it has been already been argued in

relation to markov blankets and other

important constructs in active inference

theory that the math is not the

territory

wherever you heard that one before but

this is like kind of the ultimate

instrumental

take it's it's making a clean

distinction between map and territory

between the the

utility driven use of active inference

and claims about what the system really

is

and then um

uh this just the last line but it's an

interesting quote but i'll just read the

last line is um

move to the slides

hence the description of markov blankets

could be a recurring challenge

throughout project production which may

only be resolved by reference to

decisions made in courts of law long

after the project is completed

just maybe think about i don't know

maybe like the bay bridge being built in

the bay area in california and like you

know years later and it happened

differently and i'm sure some companies

were bought and sold and

all the morphing of the organization and

the physical modification of the project

and the personnel changing over

all of that which is sort of this on one

hand it's an n equals one unique enacted

situation but then on the other hand

there's principles for engineering it

well and uh

sometimes the process is itself

wondering about how it's organized and

then sometimes you can't even know until

it's after it's done so that's just kind

of cool and funny

um

another quote

this is sort of taking a

take on philosophy from the engineering
perspective

exactly how beliefs are updated in
embodied cognition is a topic of ongoing
research which involves specialist
mathematics to describe relationships
between prior probabilities and
posterior probabilities

by contrast updating beliefs through
active inference in industrial
engineering and quality management
involves applying well-established
practical methods

in particular planning what should
happen by active inference involves
developing processes through applying
industrial engineering techniques
so it's sort of like we know what our
policy affordances and um
uh mechanism of updating our affordances
are that's our discipline
and so

more broader question is what are the
implications for generalized frameworks
like

embody cognition or active inference
what

are the implications when they interface
with disciplinary or project specific
like one-time

work

and then how does active inference uh
gets simplified in the context of
industrial engineering and how does work
in the disciplines become linked in a
bi-directional conversation to the

broader body of knowledge

just

general questions any thoughts or okay

dean yeah go for it

dean you're muted yeah i can't hear you

um

whether we're talking about the bay

bridge as

do we have the proper glove in hand

or whether we're talking about

instrumentalism do i actually have a

precise

fit of glove for my hand yeah i think i

think what what we have to be

aware of is that

both of those questions come up

post hoc

i tried on the glove and it worked or

uh maybe the design of the bay bridge

should have been different given the

the contextual circumstances

well i'm all i'm

curious about is

so before the bay bridge is built before

i i fit

one of two questions into that gloved

hand relationship am i aware of how

that can be seen post talk or ad hoc and

am i am i

am i

conscious of the fact of where i'm

taking this active inference model and

as blue pointed out earlier whose is it

and now just as importantly when

is it

when is that when is that question being

asked in the context of what the product

is because if we're going to talk about

social organizations as the product
active inference allows us to look on
both sides of that so
if as long as we're including the win in
terms of discovering fit
i think i think we're actually using
active inference
in terms of what its potential is that's
all

nice comments little philosophy like we
i'm not not not like we need that but
i'm just throwing it in there yeah
that's cool

so then here's another quote

um

in terms of the explore exploit dynamic
project production involves much more
exploration than exploitation hence more
wayfinding than navigation is involved
in particular wayfinding involves the
ability to create novel routes which are
based

on understanding a wider frame of
reference than navigating along a preset
route

wayfinding involves creating novel
routes through changing situations by
making non-conscious reference to
subjective prior knowledge and conscious
subjective reference to current
situations and semantic information

that's pretty dense but pretty
fascinating because it does point at
this kind of like repetitive
then you can just write out the
directions and follow them in a little
bit more of a rule following way

um

and then the extreme case being like a train like the caltrain just went from san francisco to san jose just went and it was like it had you know the affordance to speed up or slow down but then

when there's the bushwhacking um or the wayfinding here

then there's almost like novelty of path

and then there's higher novelties like

introducing a new tool or climbing to

the top of a tree to see further but

there's just this path finding novelty

that is uh really important for you know

all kinds of these applications so it

just

raises the question that'll be cool to

discuss

about like

how does active inference

uh juxtapose or interface with sense

making and wayfinding

explore exploit the pragmatic epistemic

distinction

and then the way that the models are fit

with accuracy minus complexity just how

does that

come together and active

any thoughts or good

yep dean

yeah because because weight finding is

my

is my niche i was just so happy that he

pulled these two

reference like if you haven't read 90

like work of 95 and 96 here i strongly

recommend it because he's pulled out two

fantastic references

in terms of uh

things you could look at in terms of
what that exploration piece actually is
that they're both fantastic especially
the 95

yeah i've read both of them and they're
both they're both excellent examples
within the context of what he's trying
to talk about in terms of implications i
just wanted to do a clap for that one
because i thought

glad glad he brought the wayfinding
piece into this cool yeah the 95 is
finding the way a critical discussion of
anthropological theories of human
spatial orientation with reference to
reindeer herders of north eastern europe
and western siberia 2009

and then the second article is humans
used predictive bayesian strategies to target
waypoints for steering 2019. so it's
kind of interesting because um
and especially having adapted a visual
foraging model
to a stigmergic pheromone deposition model
in the active inference paper
it has

drawn out this contrast between two
extremes of foraging
um both which can have a pragmatic and
an epistemic component there's like skin
in the game

ant body on the ground where you're
physically moving you're hurting and you
have to go somewhere to find out how it
is there versus visual foraging or
visual scanning is uh sort of like you
have the cost of movement of course but

it's not like um where your eye is is directly paying the cost um in most cases so it's kind of cool to see that the wayfinding and the sense making comes into play on these two extremes with a niche modifying body on the ground as well as with a scanning approach

okay um another quote

won't read the whole thing but basically we can imagine after hearing all this that the prediction error is something that you want to minimize for effective business operation so a large prediction error could lead to the company dissipating for example through its internal resources being scattered into an external state by being taken into the possession of its unplayed paid suppliers so that's um

marcotte blanket dissolution via court-mediated bankruptcy proceeding alternatively the company could be acquired and merged into one of its large unpaid suppliers that seeks to move up the value chain from tier one to oem manufacturer

in the case of acquisition and merger the description of markov blankets would need to be changed in legal documents so that just totally uh set off my scott david alarm

and

maybe think about um the ways that the markov blankets um can be designed the the

managing synthetic externalities

um

seeing law as a niche modification and
as a technology in and of itself but
also one that uh can
facilitate other technologies coming
into existence or being hindered as we
see all the time
and then
he uses the acronym bolts
business operational legal technical
social
and then the bigger question is like how
can we see active inference serve in a
de-risking or recovery capacity
and then how can we
reframe existing or emerging practices
for resilience in terms of active
inference so this is like a 1980 style
merger and acquisition
framed in the markham blanket framework
but
what does the d5
active inference model look like what
does the smart contract active inference
model look like these will be like
really interesting questions and how
they're going to intersect with formal
laws as well as the code is law
any comments
okay and then i think the last point was
just
in the further directions of the paper
further refinement of active inference
constructs is needed to enable wide
application to social organization
employing human and artificial
intelligence
apropos the practical examples in the
paper can provide starting points for

research into the challenges of
ascribing markov blankets defining
generative models handling pragmatics
and modeling variational free energy in
real world social organization and
that's an idea that we touched on in our
2020 paper active inference and behavior
engineering for teams which is that in
the case of remote teams
then
the entire structure can be observed
through the sensory data that's sent to
a user and then the actions that are
received for example on either side of
an interface or either side of a
blockchain wallet so for remote teams
because they're defined digitally we
have access to the entire state space so
to speak of sensory and active states
whereas for trying to do the um active
inference model of people in a physical
boardroom that's always going to be
complicated even if you have a really
effective way to get body position out
of the cameras in the room for example
so just kind of cool to think about
uh industrial design or engineering
all these different
ideas we talked about
in terms of online as well as physical
engineering so not just like cars and
stuff
those are hardware and software as well
so we kind of just closed by saying um
what
people can give their last thoughts blue
and dean just what would a good
understanding enable what are some good

unique predictions or implications or
areas that this

line of research might be addressing
what are the next steps for this kind of
research the goals of the research and
then also just what you're curious about
asking the author tomorrow and in the
following week

yeah i think i've laid out some
questions um today that i hope i don't
forget for tomorrow but um
i definitely think that uh

active inference in
systems and organizations and social
structures is

definitely important um along with like
the validation of your model through
other models like through the models of
other people and through like this this
confirmation bias like oh my model must
be good because you have a model that's
like my model so so i think that this
this existence um

it can help to really set boundaries and
definitions on

on um

how we really form social structures not
just businesses and and i mean in
business there's kind of a hierarchical
component and um but i mean also like
society as a whole we form these social
structures and how does this social
morphogenesis really take place so i
think really the author puts out a good
um you know way of of kind of delving
into social morphogenesis uh even
through like quality quality management
and industrial engineering but it's it's

definitely first steps

any final comments

well i'm really looking forward to

talking to the author because

i think at the end of the day rather

than

um

what's the word here rather than

exploring

why

why these these mappings have been put

together in this paper i i'm kind of

curious now about trying to figure out

so when did

when did these parallels sort of form in

his mind and get to a place where he

could take a lot of this background

information because he's i mean he's

really

cleverly and and thoroughly sourced here

when did he sort of put these two things

together as a reflection of one another

and where does he see that going forward

totally agreed it's a single author

paper that

reflects a really deep understanding and

probably a lot of experience presenting

and thinking about this and maybe even

applying it so those are all super

interesting areas to find out about

all right well blue dean thanks that was

epic thanks for all the help on the

slides and on the stream

so that was great we'll uh talk to

everybody in the dot 1.2 or later so bye

thanks dan